University of Colombo, Sri Lanka

UCSC *University of Colombo School of Computing*

BACHELOR OF SCIENCE IN INFORMATION SYSTEMS

First Year Examination - Semester I - UCSC AY22 [held in Sept./ Oct. 2025]

IS 1204 - Fundamentals of Software Engineering

(Two (2) Hours)

Answer ALL questions

64

Number of Pages = 12 Number of Questions = 4

To be completed by the candidate

Index Number: | | | | | | | | |

Important Instructions to candidates:

I. Students should answer in the medium of English language only using the space provided in this question paper.

II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.

III. Write your index number CLEARLY on each and every page of this Question paper.

IV. This paper consists of 4 questions in 12 pages (including the Cover Page).

V. Answer ALL questions.

VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.

VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate are **not allowed**.

To be completed by the examiners

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Total	

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Question 1

(a) Identify the correct System Software Type from the following list for the descriptions given in Column A and write the matching answer in Column B.

[5 Marks]

{Device Drivers, Utilities, Operating Systems, BIOS, Firmware, Programming Language Translators}

Column A - Description	Column B – System Software Type
Controls particular hardware attached to the system.	
Permanent software embedded into read-only memory.	
A translator that converts high-level language to machine code.	
Provides a platform for other software and manages hardware resources.	
Helps analyze, configure, optimize, or maintain a computer system.	

(b) Compare **Generic Software** and **Customized Software** in terms of the following aspects given below.

[5 Marks]

	Generic Software	Customized Software
Quality Concern		
Updates		
Development Cost		
Architecture		
Control		

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(c) Underline whether the following statements are TRUE or FALSE. Justify your answer in the space provided.

[10 marks]

Note: No marks will be provided, only for mentioning the statement is True/False. In either case, a proper justification is required.

i) In large systems, software costs are often greater than hardware costs. [TRUE / FALSE]

Justification:

ii) Large software is usually designed to solve wicked problems. [TRUE / FALSE]

Justification:

iii) Adding personnel to a late project often makes it faster. [TRUE / FALSE]

Justification:

iv) Use of open-source technologies or third-party tools does not affect the software development cost. [TRUE / FALSE]

Justification:

v) Poor communication among developers and clients often leads to software development problems. [TRUE / FALSE]

Justification:

(d) Select the most appropriate word phrase from the below-listed words to complete the following sentences.

[5 Marks]

{Software Engineering, Interviews, Surveys, Use-case, Requirement Elicitation, Requirement Analysis Functional requirements, Non-functional requirements, User Requirements, System Requirements, Validation, Verification, Requirement Specification document}

i) ______________________________ is the process of gathering requirements from stakeholders to define the functionalities and constraints of a software system.

ii) ___________________________ describe the behavior of a software system.

iii) ___________________________ specify the system's qualities, such as usability, reliability, and performance.

iv) The primary goal of ___________________________ is to identify and document the functional needs of the users and ensure that the system meets non-functional constraints.

v) The primary goal of gathering _____________________________ is to understand the needs and expectations of the end users for a system or product.

Question 2

(a) Identify the suitable Software Process Model from the following list for the given description in Column A and write the matching answer in Column B.

[5 Marks]

{Waterfall model, Prototype model, Spiral model, Kanban model, RAD model, Iterative model, Scrum model, Incremental model}

Column A - Type of Software Project	Column B - Software Process Model
Emphasizes rapid development with component reusability and prototyping, often using automated tools	
Breaks the project into smaller deliverables, each functioning independently but building towards the complete system	
A working version of the system is built quickly to understand requirements before full-scale development	
Combines iterative development with risk analysis, moving in loops to handle uncertainties.	
Delivers the system in repeated cycles, allowing improvements based on feedback in each cycle.	

(b) Consider the following definitions related to the SCRUM terminology.

(A). The sole person responsible for managing the Product Backlog.
(B). A dynamic list of items that may be added, removed, or reprioritized based on the progress.
(C). A development iteration that is usually 2 to 4 weeks long.
(D). A daily meeting of the Scrum team that reviews progress and prioritizes work to be done on that day.
(E). Responsible for delivering the work through the sprint.
(F). A piece of work that is potentially shippable.
(G). A collaborative meeting takes place at the end of each iteration to focus on the product and its functionality.
(H). Responsible for interfacing with the rest of the company and ensuring the Scrum team is not diverted by outside interference.
(I). A collaborative meeting held at the beginning of each iteration to define the work that will be undertaken during the iteration and to set clear goals and expectations.
(J). An inspect-and-adapt cycle to enable the team to continuously improve its processes and performance incrementally over time.
(K). Ordered list of all features, functions, requirements, enhancements, and fixes that constitute the changes to be made to the product in future releases.

Identify the appropriate SCRUM terminology for the above definitions and categorize them into Roles, Artifacts, and Ceremonies within the SCRUM Framework. Specify the Letter of each statement and the relevant SCRUM terminology in the appropriate space.

For example: [A] - Product Owner

[10 marks]

SCRUM ROLES	SCRUM ARTIFACTS	SCRUM CEREMONIES
[A] - Product Owner		

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(c) Identify the suitable software requirement checking aspects from the following list for the given description in Column A and write the matching answer in Column B.

[5 Marks]

{Adaptability, Verifiability, Validity, Completeness, Traceability, Consistency, Comprehensibility, Realism}

Column A - Description	**Column B – Requirement checking aspect**
Ensures the requirements accurately reflect the customer's real needs.	
Ensures that no two requirements contradict each other.	
Ensures all necessary requirements for the system are specified.	
Ensures the requirements can be implemented within constraints of technology, budget, and time.	
Ensures each requirement can be tested or demonstrated to confirm it has been met.	

(d) List and briefly explain **five (5)** common reasons why software requirements may change during the development process.

[5 Marks]

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Question 3

(a) For each description provided in ***Column A***, identify the correct name of the corresponding object or relationship used in a ***Use Case Diagram*** and write it in ***Column B***. Then, illustrate the appropriate ***UML symbol*** for it in ***Column C***.

[5 Marks]

Column A - Description	**Column B – Object/Relationship name**	**Column C- Symbol**
Represents a functionality or service provided by the system.		
Represents the scope of the use case.		
Represents a user or any external system that interacts with the system.		
Represents an optional or conditional relationship between two use cases.		
Represents a compulsory relationship where one use case always uses another.		

(b) Identify the suitable word from the following list for the given descriptions regarding the class diagrams in Column A and write the matching answer in Column B.

[5 Marks]

{Attribute, Aggregation, Abstract class, Association, Class, Composition, Generalization, Multiplicity, Operation, Role}

Column A - Description	**Column B - Answer**
A property or characteristic of a class that holds data.	
A structural relationship that shows how two classes are connected.	
An inheritance relationship where a one class shares attributes and operations of a another class.	
The purpose or behavior a class plays in an association.	
A class that cannot be instantiated and often defines common behavior for its subclasses.	

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(c) Consider the following sequence of messages with regards to an ATM cash withdrawal process.

1. **Customer → ATM**: *insertCard()* → Creation Message (ATM session created).
2. **ATM → Customer**: *enterPIN()* (Synchronous message).
3. **ATM → Bank Server**: *verifyPIN()* (Synchronous message).
4. **Bank Server → ATM**: *PINVerified / PINInvalid* (Return message).
5. **ATM → Customer**: *selectWithdrawal(amount)* (Synchronous message).
6. **ATM → Bank Server**: *requestTransaction(amount)* (Synchronous message).
7. **Bank Server → Account**: *debit(amount)* (Synchronous message)
8. **Account → Account:** *validateBalance()* (Reflexive message).
9. **Account → Bank Server**: *transactionSuccess()* (Return message).
10. **Bank Server → ATM**: *approveWithdrawal()* (Return message).
11. **ATM → Customer**: *dispenseCash()* (Asynchronous message).
12. **ATM → Customer**: *printReceipt()* (Asynchronous message).
13. **Customer → ATM**: *endSession()* → **Destruction Message** (ATM session ends).

Draw the corresponding sequence diagram. The diagram should reflct all the relevant components of a sequence diagram to obtain full marks (eg: actors, objects, lifelines, activation bars, different types of messages, etc.).

[10 makrs]

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(d) Identify the suitable Software Architectural Pattern form the following list for the given scenarios in column A and write the matching answer in Column B.

{MVC architecture, Layered architecture, Repository architecture, Client-Server architecture, Pipe & Filter architecture}

[5 Marks]

Column A - Scenario	**Column B – Software Architectural Pattern**
Banking software systems - where user interfaces, business rules for transactions, data validation logic, and database operations should be organized in distinct to ensure security, maintainability, and regulatory compliance.	
Online multiplayer games - where game users on players' devices communicate with centralized game servers that maintain the authoritative game state, handle player interactions, and synchronize gameplay across all connected players.	
Social media platforms like Facebook - where user profiles and posts are displayed through various interfaces while user interactions like posting, commenting, and sharing are handled by business logic controllers.	
Image processing applications like Photoshop - where raw image data flows through a series of independent filters (blur, sharpen, color correction, resize) that can be combined and reordered to create complex image transformations.	
Digital library management systems - where books, research papers, and multimedia content are stored centrally with standardized access methods for cataloging, searching, and retrieval by librarians, students, and researchers.	

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Question 4

(a) Select the most appropriate Object-Oriented Concept from the following list for the given description. Write the selected answer in the provided space.

{Object, Abstraction, Interface, Encapsulation, Method Overriding, Inheritance, Polymorphism, Composition}

[5 marks]

i) The mechanism where a new class can inherit properties and methods from an existing class.

____

ii) The ability for objects of different classes to respond to the same method call in their own specific way.

____

iii) A concrete instance of a class that contains actual data values and can perform the behaviors defined by its class blueprint.

____

iv) The process of simplifying complex systems by hiding unnecessary implementation details and exposing only the essential features that users need to interact with.

____

v) The practice of bundling data and methods together within a class while restricting direct access to internal details, providing controlled access through public interfaces.

____

(b) In software implementation, describe **two (2)** main techniques that can be used to increase the reliability of the code.

[4 Marks]

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(c) Identify the suitable software testing type from the following list for the given descriptions in Column A and write the matching answer in Column B.

[5 Marks]

{Unit testing, Integration testing, Load testing, System testing, Acceptance testing, Stress testing, Regression testing, Performance testing, Usability testing, Security testing}

Column A – Description	Column B – Test
Testing system behavior under extreme conditions beyond normal capacity to identify breaking points and failure modes.	
Testing the user experience to ensure the software is intuitive, easy to use, and meets user expectations for functionality and design.	
Testing system performance under expected normal operational conditions to verify it can handle the anticipated number of users.	
Testing performed to determine whether a system meets business requirements and is ready for delivery to end users.	
Re-running previously passed tests to ensure that new code changes haven't introduced bugs or broken existing functionality.	

(d) Fill in the blanks with the most appropriate word phrase to complete the following sentences about software testing.

[5 makrs]

i) When a nearly finished product is released to a limited group of real-world users to gather feedback under actual usage conditions, this is known as ____________________________.

ii) When users of the software work with the development team in a controlled environment to catch defects before releasing to external users, it is called ___________________________.

iii) Before a system goes live, clients verify whether the software meets their business needs and contract requirements through ___________________________.

iv) A testing method where the tester examines paths, loops, and logic within the source code is known as ___________________________.

v) A testing approach where the tester checks the system's functionality by giving inputs and evaluating outputs, without knowing the internal code, is called ________________________.

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(e) List down and describe **three (3)** main types of software maintenance. Give an example for each. [6 marks]

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